THEOREM NAGATA

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter spaces-pushouts.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the document wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate was previously judged H2 by the initial-corpus stage; that judgement and its reason are reused verbatim and were not re-derived here. Original candidate key: S001-spaces-pushouts-theorem-nagata.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original section 03I3: sources/spaces.tex, source lines 2719–2719. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original reference 0873: sources/spaces-limits.tex, source lines 2874–2874. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 07SU: sources/spaces-limits.tex, source lines 2111–2111. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original reference 0851: sources/spaces-limits.tex, source lines 1694–1694. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 0463: sources/spaces-morphisms.tex, source lines 5361–5361. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 082K: sources/spaces-more-morphisms.tex, source lines 7755–7755. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 04NZ: sources/spaces-morphisms.tex, source lines 9270–9270. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 04XY: sources/spaces-morphisms.tex, source lines 7928–7928. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 04WP: sources/spaces-morphisms.tex, source lines 7916–7916. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original reference 06NH: sources/spaces-properties.tex, source lines 1647–1647. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BA8: sources/properties.tex, source lines 523–523. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01ZV: sources/properties.tex, source lines 5141–5141. Navigation pointer used in the counted span; the span contains the pointer

only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 0F41: sources/flat.tex, source lines 9950–9950. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F4C: sources/spaces-pushouts.tex, source lines 3541–3541. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0F4B: sources/spaces-pushouts.tex, source lines 3276–3276. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Theorem 0.1. *Let S be a scheme. Let B be a quasi-compact and quasi-separated algebraic space over S. Let $X \to B$ be a separated, finite type morphism. Then X has a compactification over B.*

Proof. We first reduce to the Noetherian case. We strongly urge the reader to skip this paragraph. First, we may replace S by $\mathrm{Spec}(\mathbf{Z})$. See Spaces, Section 03I3 and Properties of Spaces, Definition 03BS. There exists a closed immersion $X \to X'$ with $X' \to B$ of finite presentation and separated. See Limits of Spaces, Proposition 0873. If we find a compactification of X' over B, then taking the scheme theoretic closure of X in this will give a compactification of X over B. Thus we may assume $X \to B$ is separated and of finite presentation. We may write $B = \lim B_i$ as a directed limit of a system of Noetherian algebraic spaces of finite type over $\mathrm{Spec}(\mathbf{Z})$ with affine transition morphisms. See Limits of Spaces, Proposition 07SU. We can choose an i and a morphism $X_i \to B_i$ of finite presentation whose base change to B is $X \to B$, see Limits of Spaces, Lemma 07SK. After increasing i we may assume $X_i \to B_i$ is separated, see Limits of Spaces, Lemma 0851. If we can find a compactification of X_i over B_i, then the base change of this to B will be a compactification of X over B. This reduces us to the case discussed in the next paragraph.

Assume B is of finite type over $\mathbf{Z}$ in addition to being quasi-compact and quasi-separated. Let $U \to X$ be an étale morphism of algebraic spaces such that U has a compactification Y over $\mathrm{Spec}(\mathbf{Z})$. The morphism

$$U \longrightarrow B \times_{\mathrm{Spec}(\mathbf{Z})} Y$$

is separated and quasi-finite by Morphisms of Spaces, Lemma 0463 (the displayed morphism factors into an immersion hence is a monomorphism). Hence by Zariski's main theorem (More on Morphisms of Spaces, Lemma 082K) there is an open immersion of U into an algebraic space Y' finite over $B \times_{\mathrm{Spec}(\mathbf{Z})} Y$. Then $Y' \to B$ is proper as the composition $Y' \to B \times_{\mathrm{Spec}(\mathbf{Z})} Y \to B$ of two proper morphisms (use Morphisms of Spaces, Lemmas 04NZ, 04XY, and 04WP). We conclude that U has a compactification over B.

There is a dense open subspace $U \subset X$ which is a scheme. (Properties of Spaces, Proposition 06NH). In fact, we may choose U to be an affine scheme (Properties,

Lemmas 0BA8 and 01ZV). Thus U has a compactification over $\mathrm{Spec}(\mathbf{Z})$; this is easily shown directly but also follows from the theorem for schemes, see More on Flatness, Theorem 0F41. By the previous paragraph U has a compactification over B. By Noetherian induction we can find a maximal dense open subspace $U \subset X$ which has a compactification over B. We will show that the assumption that $U \neq X$ leads to a contradiction. Namely, by Lemma 0F4C we can find a strictly larger open $U \subset W \subset X$ and a distinguished square $(U \subset W, f : V \to W)$ with V affine and $U \times_W V$ dense image in U. Since V is affine, as before it has a compactification over B. Hence Lemma 0F4B applies to show that W has a compactification over B which is the desired contradiction. $\square$